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RRAC: Role based reputed access control method for mitigating malicious impact in intelligent IoT platforms

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ABSTRACT

IoT Environment is a recent development in communication networks that supports a wide range of individual and cooperative user applications. Due to the openness of the service providing network, communication services are exposed to malicious risks. In this manuscript, a role-based reputed access control (RRAC) method is designed to mitigate malicious attacks in IoT environments. The RRAC operates in two categories: providing an adaptive certificate authentication internally and trusted user communication externally. The internal security method is responsible for authenticating communication between users and resources. The external security feature aids secure service discovery and user selection for discovering services. The integrated security methods mitigate the influence of malicious users and forged resources in the network, improving communication reliability. The access control decides the availability and usability of the devices to participate in service communications either as a neighbor or as a resource. Experimental results show that the proposed RRAC improves the performance of IoT device communication by achieving higher detection and minimizing detection time, service overhead and loss, false positive rate and misdetection.

1. Introduction

IoT environment is a collection of end-users, wireless mobile devices, computational and interactional machines, applications and networks that exploit information in a smart manner. This environment adapts with the user demands in the smart world with intelligent communication and decision making features [1,2]. The perspective of an intelligent communication environment is to develop smart applications that are user-friendly to satisfy the user demands [3,4]. The smart applications integrate multi-level decision making, sharing and access control, resource allocation tasks that are expected in the smart world for data processing and information sharing communications. Improved reliability of the devices and their interoperability towards other heterogeneous platform provides a range of benefits including simple and robust resource utilization, diminished physical workspace, pervasive information access, efficient sharing and secure service utilization [5,6].

IoT environment adapts to heterogeneous networks, communication devices, wireless technologies, a multi-level resource for easing end-user request processing. Recent developments of machine-to-machine interaction and Internet-of-Things (IoT) focus to extend the functions of the IoT environment to the edge network to satisfy service demands of the user [5]. As access, processing and retrieval of information are integrated with the real-world entities, the IoT platform provides seamless service response for the end-users. With the extension of communication technology, the interoperable feature of the IoT environment permits small-scale network integration with large-scale clouds for information sharing [7,8]. The integration of large-scale cloud with the IoT users or small-scale networks demands appropriate platform, technology and privacy solutions. Users expect a secure and privacy-preserving resource/ information sharing in interoperable networks where the density of users and resources is high. The characteristics of cloud platforms such as heterogeneity, dynamicity, flexibility are inherited directly in IoT environments [9,10].

In an IoT environment, request processing and resource availability occur outside the end-user platform. The service provider (SP) whose resources are distributed in cloud lends resources, applications and services as a response for the end-user request. The SP interacts with the IoT environment either directly or through recommended networks and infrastructures for serving the user request. Therefore, the dependability factor is extended behind the SP where heterogeneous networks and devices are available [10,11]. The SP grants dynamic scheduled resources that support real-time services in a seamless manner. Communication interruption due to denial of service (DoS) or the presence

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of adversaries in the heterogeneous environment degrades the quality of service (QoS) and the end-user demands remain un-serviced. Handling delayed service response; outage communication and resource unavailability are the primary challenges in interoperable features of the network [12,13]. As the services are shared across heterogeneous networks, privacy and security of the resource and the sharing user/SP is the next challenging task in QoS [14].

For real-time service response, application and user-level security are prominent in achieving better experience of information sharing and resource access. Authentication methods and cryptography-based security measures provide defense against vulnerabilities in the interoperating network [15]. The IoT environment integrates security mechanisms for mitigating the impact of security issues in user interactions. IoT environment is restricted to collaborative authentication and security schemes due to its openness and flexible operating nature [16, 17]. The contributions of the paper are two-fold:

(a) Designing a reputation based neighbor access for ensuring device security in an IoT environment. This access control determines the role of the device as a forwarder or a resource. Besides, security is achieved by selecting reliable neighbors preventing selfish devices participating in the communication.
(b) Designing an authentication method for securing device-to-device communications. The authentication method relies on hyperelliptic cryptography to ensure communication is secured between the devices.

The above contributions are integrated to achieve fair security in IoT device communications and a comparative study is introduced to verify the consistency of the proposed method.

The remainder of the paper is organized as follows: Section 2 briefs the related works that have been proposed in the past with their pros and cons. Section 3 discusses the proposed methodology with the explanation on device and communication security. In Section 4, the performance analysis of the proposed method is discussed with the simulation settings and appropriate comparisons followed by a conclusion in Section 5.

2. Related works

Han et al. [18] proposed a three-layer network structure for improving communication in a congested IoT environment. The nodes in the communication environment are grouped into clusters to detect the congestion in the network. The communication and region identification process is carried out independently to minimize message overhead.

Lin and Tang [19] introduced a blockchain-based authentication scheme for improving device-to-device communication reliability. The authentication of channel state information aids the accessibility features of the end-users. Cross-tier interference employed also assists in counterfeited user detection and access. This authentication scheme improves the spectral efficiency of device-to-device communication.

Fog-assisted software-defined networking based intrusion detection/ prevention system (FSDN-IDPS) is proposed by Shafi et al. [20] for detecting anomalies in IoT networks. This detection/prevention system is more consistent in detecting real-time attacks with the help of SDN. Besides this system also provides an optimal solution for allocating fog resources in resource-constrained IoT networks. This system improves detection accuracy and minimizes false positives. This system is unreliable in communication authentication at the time of service responses.

Novo [21] proposed access management architecture for IoT user communication. This architecture is constructed using blockchain technology that is best suitable for scalable real-time IoT networks. This architecture improves network throughput by improving the transmission of messages.

Proxy-aided cipher text-policy is designed by Cui et al. [22] for improving access control of the data shared in edge computing networks. The encryption process is distributed among the available edge devices to minimize computation time. Different from the conventional attribute-based encryption, this method does not require secure channels for key distribution.

Dynamic access control (DAC) mechanism is introduced by Jang et al. [23] for improving the efficiency of group paging employed IoT networks. This mechanism provides adjustable random access based on available physical channels and preambles. The random access performance is augmented by integrating a collision detection technique. This mechanism improves resource, energy and transmission efficiency by minimizing paging control time.

Mavropoulos et al. [24] discussed the IoT security framework "Apparatus" with the modifications in its structural language. The modifications are carried out for extending the support of the security for public transport systems. The modifications in the structural language are monitored and modeled using class-based notation modeling for unanimous operations of "Apparatus" in both design and implementation phases.

Yan et al. [25] proposed a node authentication scheme for improving the efficiency of privacy-preserving in pervasive social networking. In this scheme, the trust of the nodes is updated by multiple authorized parties facilitating the communication features. This method achieves less computational complexity and communication cost through cooperative trust-based authentication.

Akyildiz and Kak [26] introduced the internet of space things (IoST) for achieving cost-effective connectivity between 5G users. IoST is a cyber-physical system that is designed for ensuring fine-grain access over the hardware and software resources of the network. With the help of software-defined networking and network virtualization functions, IoST improve network resource utilization with less operational costs.

SecTrust routing protocol is proposed by Airehrour et al. [27] for minimizing packet loss and improving the rate of attack detection in low-power lossy networks (SecTrust-RPL). This routing protocol is efficient against Sybil and rank attacks by detecting and mitigating the adversaries based on trust. This routing protocol considers the direct and recommendation trust through a series of observations to update the trust value. This method takes additional detection time to identify the adversaries due to the classification process.

Li et al. [28] proposed a heterogeneous deniable authentication for securing communication in pervasive environments. The heterogeneous authentication comprises of two protocols for permitting a public key used to communicate with an identity-based cryptography employing user and vice versa. The verification of the authenticating users is progressed through batch verification supported by the authentication process. As the protocol is of two types, this method consumes less time for authentication.

Eom and Huh [29] introduced a novel restrictive link-ability property for privacy-preserving in ubiquitous communication environments. The link-ability property is achieved using a group signature that enables encryption and decryption using the same pseudonym.

Yuan and Li [30] proposed a multi-source feedback trust information fusion mechanism (MFTM) for improving the security of IoT devices. MFTM is a light-weight trust mechanism that operates on the cooperation between the IoT nodes. The initial device and trust configuration for communication facilitates the cooperative feature of the nodes to receive feedback upon which the trust is updated. The setback of the mechanism is its prolonged configuring time and uncontrolled service loss.

Hammi et al. [31] proposed a bubble of trust scheme for securing IoT communications based on trust. This scheme is based on blockchain authentication distributed among the available devices grouped in virtual regions. This virtual grouping ensures the integrity and the availability of communicating data that is dependent on the trust of the device.

From the above survey, the general issue in the existing methods is its lack of providing dual authentication schemes i.e. device-level and communication-level security. More precisely, the methods discussed in [20,27] and [30] are vulnerable to service authentication, timely response and single-level trust management respectively. Some other methods require complex computations [22] or support a small level of communication [19]. Considering these issues, in this manuscript both device and communication securities with less detection time and service loss are introduced.

3. Methodology

The proposed RRAC is designed to ensure security for both inter and intra IoT device user communication. In an IoT environment, the immediate neighbor for communication can be either infrastructure or a like device. The medium of communication is wireless, an interruption between users defaces the quality of information exchange. The presence of an adversary inside and outside the IoT communication environment degrades resource and information sharing. Communication loss, DoS, resource unavailability and resource exploitation are some of the issues due to the presence of adversary. Fig. 1 represents an IoT communication environment model interacting with the cloud network.

IoT Environment: The IoT environment is self-replenished with resources and wireless technologies for facilitating inter and intra user communication. It connects with a variety of heterogeneous networks such as cloud and ad-hoc in a distributed manner with the help of wireless technologies the users and devices in IoT environment extend their access and sharing requests to other heterogeneous networks IoT environment consist of intelligent self-decision making devices that help to meet user requirements.

Infrastructure: The infrastructure consists of base stations, access points and other components that facilities interaction between heterogeneous users. The infrastructure wraps a wide geographical distance that enables distributed resource access. Infrastructure elements are service concentrated and distributed to handle user requests and thereby organizing communication.

Cloud Network: Cloud Network serves as a repository for storing information and services. Cloud network interconnects a wide range of servers, storage, applications and services. The cloud is a pervasive environment capable of serving user demands through information access permission or leasing its services. Cloud is accessed through a common network like the internet and deployed infrastructures. SP is responsible for lending applications and functionalities across the cloud for the request generating user.

The proposed RRAC consists of two-levels of security: device security and communication authentication. The role of the IoT device user is of two types: resource provider or service handler. This role is determined using the reputation estimated in its device level and service level security offered. The individual process of security is explained briefly in the following sections.

3.1. Device security

The end-user device initiates a request for service access to the cloud network. At the initial stage, the cloud SP registers the information of the user. The information such as device identifier, physical address and request time are registered by the cloud service provider. The SP generates authentication certificates with information such as (ID, UCN, TS, DS, V CSN, AC). The tuples of the authentication certificate are explained in Table 1.

The certificate possessing device is alone capable of participating in the communication. If the device owns an outdated certificate or any discrepancy is found in the certificate, the user is turned malicious. Other users communicating with the device post verification perform operations at their own risk.

Table 1
Authentication certificate tuples and description.

Tuple	Description
ID	Unique identification number of the device
UCN	Unique certificate series number (of the same service provider)
TS	Time stamp that represents the validity of the certificate
DS	Digital signature as recommended by the service provider
V	Version of the certificate
CSN	Certificate provider name
AC	Access count of the device for multiple service providers

The SP is capable of varying the TS of a leased certificate or revoke it depending upon the usability of the certificate by a device depends on the time and service span as requested and accessed by the user device. The behavior of the device is characterized by its repletion and information exchange model exhibited with the certificate. The reputation of a device (R_d) is estimated using two methods response from the neighbors and characteristics of information transfer at the time of communication.

Let d_n represent the number of devices excluding the infrastructure with which a source device has communicated until a time span t_s. The reputation is modeled using Eq. (1)

$$R_d = \sum_{i=1}^{d_n} \rho_{(s,i)} \times c_r, \tag{1}$$

where c_r is the communication reliability that is given as

$$c_r = \frac{c_s - c_f}{c_t}, \tag{2}$$

where c_s, c_f and c_t are the successful, failure and total communication expedited respectively by the device d. Here, $\rho_{(s,i)}$ is the probability of connection truth between the source device s and ith device $\in d_n$. $\rho_{(s,i)} = 1$ If s has communicated with i at least once else $\rho_{(s,i)} = 0$. The reputation of d is estimated for devices that own SP certificates and utilize the same certificate for communication. The SP holds an activity record of the certificate exploiting device with its R_d to ensure secure device availability. The SP updates the reputation with shared UCN and ID of the device. Unlike the conventional methods [25], the SP does not account R_d directly. Contrarily, it updates the reputation with the variation constraint (ΔE). The variation constraint is an exception that is updated at a time instance t. The error is variation constraint is estimated using Eq. (3)

$$\Delta E = \left| R_d(t) - \sum_{i=1}^{d_n} \frac{R_{d_i}(t-1)}{d_n} \right|. \tag{3}$$

The SP saves the activity record of an d with the tuples $\left[ID, UCN, t\left(R_d - \Delta E_t\right)\right]$. The record holds the values of $\left(R_d - \Delta E_t\right)$ for a certain threshold R_{th} is not updated and the certificate is revoked. The R_{th} is estimated using Eq. (4)

$$R_{th} = \left| \left[\frac{d_r.R_r + d_{nr}.R_{nr}}{(d_r + d_{nr})} \right] - R_{nr} \right|, \tag{4}$$

where d_r and d_{nr} are the counts of reputed and not reputed devices respectively and R_r and R_{nr} are the expected reliability values from d_r and d_{nr} respectively. After estimating R_{th}, the reputation update is filtered for $\frac{d_r}{(d_r+d_{nr})}$ devices in the IoT environment. The final reputation update ($\prod R_d$) is computed as

$$\prod R_d = \sum_{i=1}^{d_n} \frac{R_d - \Delta E_t}{d_r}. \tag{5}$$

Eq. (5) holds for a new device update by the SP. Contrarily, if the same device is to be updated with its recent reputation observed after a time span t_s is given as

$$\prod R_d^* = \left[\beta * \prod R_d + (1-\beta) \prod R_d\right] - \left[\Delta E_t / t_s - t\right], \tag{6}$$

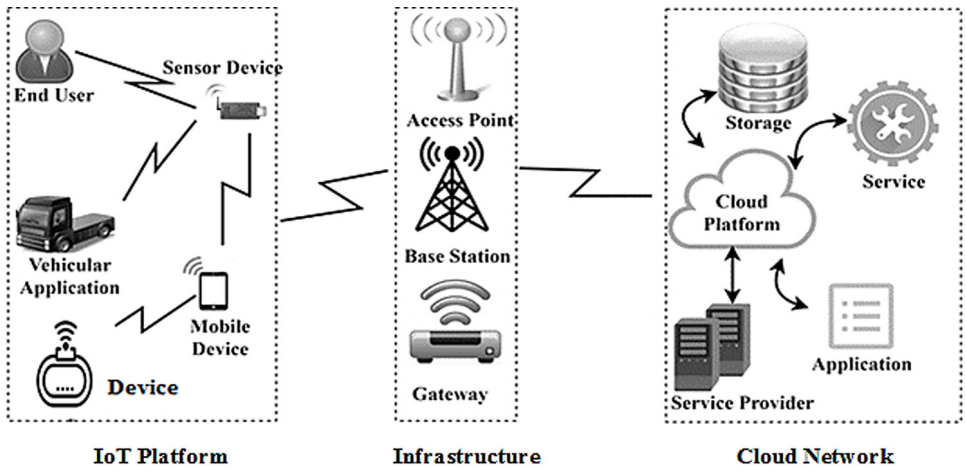

Fig. 1. IoT communication environment model.

where $(t_s - t)$ represents the time difference between the last update time t and the current time span t_s. β is a constant related to the communication of the devices. β is maximum for all $\rho_{(s,i)} = 1$ i.e. the communication constant is maximum if there exists a communication path between the device s and device i.

The updated reputation is held by the SP that ensures the reliability of the communicating device. A device requesting communicating or resource access from another device located in different pervasive IoT environment is notified by the SP for its reputation information. Device registration and reputation update are replicated to the distributed SP for gaining control over the user activities in different places and with different service requests to a new SP. A security administrator or a SP management tracks the activities of the device based on location, service access and sharing with the AC update. For ease of device verification, an IoT device confirms the status of the neighbor or service provider. The status of the certificate as an update by the SP is used to verify the reputation more instantly. In the application layer of the IoT device user, the security risk in accessing a service from a reputation updated resource device is notified. The notification carrier the message as high, medium or low based on the risk. For a device (resource (neighbor) with less $\prod R_d^*$, the security risk is high. The device with $\prod R_d^* < R_{th}$, the certificate is revoked and AC is decreased by one.

The process of certificate revocation is a special feature of the SP to mitigate distributed access of malicious or adversary users/devices. The SP monitoring the activities of the device constructs a reputation index R_i post $\prod R_d^*$ update. The reputation index decides the reliability of the device in terms of security. The reputation index is estimated using Eq. (7)

$$R_i = 1 - \frac{1}{d_n} * R_r, R_r \in \left\{ \prod R_d < R_{th} \right\}. \tag{7}$$

Eq. (7) précises neighbor/device exclusion based on the reputation value and available device that already update R_r. The certificate revocation metric $\nabla\varphi_r$ is then computed as

$$\nabla\varphi_r = \sum_{i=1}^{d_n} \rho_{(s,i)} * \left[R_i - \frac{1}{d_n} * \left| R_{nr} - R_r \right| \right]. \tag{8}$$

The $\nabla\varphi_r$ takes the range $[0, 1]$ such that if $\nabla\varphi_r > R_{th}$ then the certificate is revoked. The revocation conditions are, therefore, $\prod R_d < R_{th}$ or $\prod R_d > R_{th}$. The diverse conditions enable detection of the adversary with the consideration of the impact of varying d_{nr} and R_{nr} density. The above reputation estimation is resilient against the density of malicious users. DoS due to the presence of several malicious users and selfishly behaving adversaries are identified by the reputation management process. Fig. 2 illustrates the reputation management.

The role and functions of the device in an IoT environment are determined post the reputation update process. The participating role

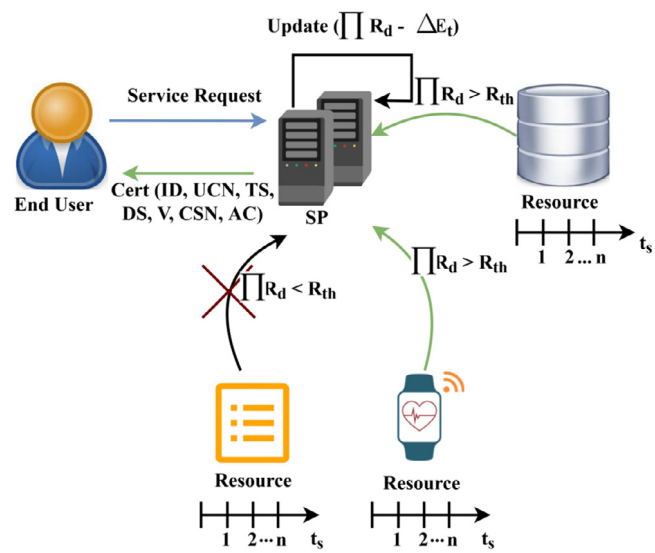

Fig. 2. Reputation management process.

and communication functions are enabled only when $\nabla\varphi_r < R_{th}$ verified by means of d_r and R_r from the available d_n. The process of update is done at any time instance t for a time span t_s. The reputation management is seamless until the TS of the certificate as allocated by the SP. The timestamp of an unreliable or reputation less neighbor is the instance time t in t_s.

In another context, if the service time of a resource exceeds the allocated TS of the certificate, the user sends a request to the SP. The SP updates the TS of the certificate until the service closure time. Let s_{sd} and s_{cd} represent the service start time and closure time, if $(s_{sd} + s_{cd}) > TS$ then

$$TS = TS + (TS - S_{cd}) = 2TS - S_{cd} \tag{9}$$

is the extended TS in the certificate. Similarly, reputation management is extended for an additional slot covering $(2TS - S_{cd})$. The available d_n is subjected to change in the extended time period. SP extends the TS of an user/device only if $\nabla\varphi_r < R_{th}$. This reputation decides the role of participation or denial in the user communication between devices externally.

3.2. Communication security

In the communication security process, Service-level authentication is provided for the requests and resource response. The authentication

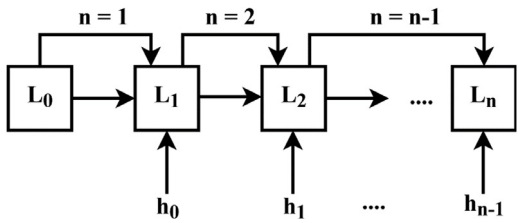

Fig. 3(a). Continuous communication hash.

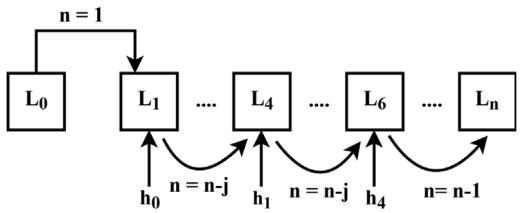

Fig. 3(b). Interrupted communication hash.

process is enabled through a linear chained authentication signature. The communication of the devices with a higher reputation or $\nabla\varphi_r < R_{th}$. The authentication as recommended in the certificate dispatched by the SP is used public-key security. This authentication scheme is collaborative and interoperable between devices connected to the same or different SPs. Therefore, the device level private key authentication is facilitated using linear chained authentication. Linear chained hashing is based on TESLA [32] for supporting sequential key generation for authentication. The process of authentication is of two stages: hash generation and signature-based authentication.

The process of communication authentication begins with the initialization of authenticating the value generation. Let L represent the communication sequence observed between two successive time slots t_s. The sequence is divided into smaller chunks (i.e) $L = \{L_1, L_2, \ldots, L_n\}$ depending upon the availability of the neighbor. Each L retains an individual time period $t_p = \{t_{p_1}, t_{p_2}, \ldots, t_{p_n}\}$. The initial L (i.e) L_o is the hash generating sequence for all the L in t_p. By the property of TESLA [19], the previous hash of L is identified using the most recent L; the sequence L_i in L is thus estimated as

$$L_i = \mathrm{h}\left(L_{i+1}\right), i \in [0, n-1]. \tag{10}$$

In this manuscript, the process of double hashing as in conventional TESLA is not used to generate the key. As an alternative method, the last updated $\prod R_d$ is used for generating the hash and hyper-elliptic cryptography is used for signature generation.

3.2.1. Hash generation

In a linear hash generation, the previous L_i is used for the hash functions. As hash for the ith sequence of L is generated using Eq. (11)

$$h_i = R_n * L\left|2^{n+j}\right|, j = 0, 1, \ldots < (n-1) \tag{11}$$

where R_n is a random generator used to avoid replicated hash. The process of hash generation is linear with no backoff and replacement. This coincides with the separated L mapped to t_p. The above hash is dependent upon L that is divided into t_p. For each L_i, the hash is estimated using Eq. (11) for its previous state. The process of hash generation is illustrated in Fig. 3(a) for continuous and interrupted communication.

For a continuous communication, n value is $n + j = \{1, 2, \ldots, n\}$ where the previous state od L is retained for hash generation (i.e) $\{h_0, h_1, h_2, \ldots, h_{n-1}\}$ for $\{L_1, L_2, \ldots, L_n\}$. This is most precisely sequential each that the number of hash generated h_g for n chunks is

$$h_g(n) = n\left(\frac{1+n}{2}\right) + (1-n) \tag{12}$$

$$h_g(n) = \frac{n^2 - n + 2}{2} \tag{13}$$

The communication for n splitter L requires $h_g(n)$ for all $(n+j)$ and therefore, R_n is generated until $(n+j)$. The $(n+j)$ takes an additional time t_{p_n} for which the TS is extended and hence, $t_{p_n} = t_{p_n} + s_{cd}$. In an interrupted communication (due to reputation update), the number of $h_g(n)$ is less within the given t_{p_n} or $(t_{p_n} + s_{cd})$. However, the previous state of the hash is retained for all current L, (i.e) if the sequence is non-linear, then the last generator hash $(n-j)$ is retained for nth the L.

If the interrupted hash exceeds $(t_{p_n} + s_{cd})$ then the communication is unsuccessful in determining the reputation of the device.

3.2.2. Signature-based authentication

The communication between the devices is secured by employing digital signature authentication. The digital signature is generated using conventional hyperelliptic curve cryptography. Let $C: y^2 + pf(x)y = f(x)$ represent a hyperelliptic curve with a polynomial function $pf(x)$ and y is an arbitrary point on the curve. The arbitrary point y of C is assumed to be present in a region $E[pf(x)y]$. The signature for authentication is constructed by selecting a random y points as

$$y = g^x |p|, \tag{14}$$

where g and p are the start value of L and the largest prime number in $pf(x)y$. With the R_n used in the hash generation, a range exponent value (r) is generated using Eq. (15) and signature as is generated using Eq. (16)

$$r = g^{R_n} |p|, \tag{15}$$

where g is a random generator for the range of p prime numbers located in y of the hyperbola.

$$as = \frac{1}{R_n}(sm - xr)|p|, \tag{16}$$

where sm is the service message of sequence L.

The communicating device shares (g, y, p) with the destination device. Then the authenticated message (am) is generated on verifying the $\prod R_d$ of the destination device.

$$am = \left(as.h_i\right) : \left(L_i, t_{p_i}\right). \tag{17}$$

The generated message is transmitted to the destination/source device. As R_n is attached to the as, unlike the existing authentication schemes, the random value need not be attached to the am.

For each transmission sequence $\{L_1, L_2, \ldots, L_n\}$ the time slot $\{t_{p_1}, t_{p_2}, \ldots, t_{p_n}\}$ is mapped. The integrity of the transmitted service message is verified by

$$\left\{t_{p_1} \cap L_1 \cap as_i\right\} = \begin{cases} h_{i-1}, & for\ continuous\ communication \\ h_{n-j}, & for\ interrupted\ communication \end{cases}. \tag{18}$$

In Eq. (18), the message integrity at the receiving device is verified in two-cases: continuous and interrupted communication (Figs. 3(a) and 3(b)). For continuous communication, the hash for $(i-1)$ path devices is to be estimated as the message hops $(i-1)$ devices. On the other hand, in an interruption, the communication resuming time is not known and hence the hash is estimated from the resuming (n–j) device.

If there is a discrepancy in the integrity process, then the service is disconnected and $\prod R_d^*$ is updated. If the service response is negative, the reputation of a device decreases. For a successful communication,

$$am + R_n - \left(R_n * g\right) = as + R_n(g) - \left(R_n * g\right) = am. \tag{19}$$

The authenticated message and the reputation for the generated sequence g are verified with the authentication signature and the reputation of the nth device to get the authenticated message. In this process, the reputation of the device n is considered at the transmitting and receiving end with the generator g such that the final translation results in an authenticated message alone. This verifies the response

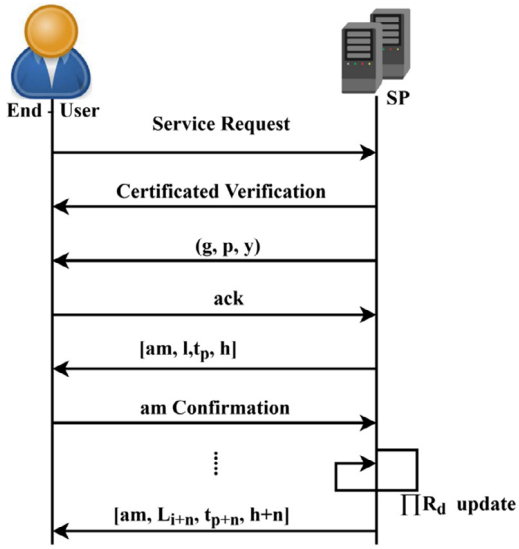

Fig. 4. Service communication authentication.

of the service to the requesting user. The above process is designed for reputed devices in the IoT communication environment. The role of the device as a resource or a request processing neighbor is determined by the *as* and $\prod R_d$ factors respectively. With the estimation of these factors, the availability and usability of the devices are determined. The above authentication method ensures secure communication between the devices. The *am* is decoded at the receiver using a precise signature. Adversary devices that do not possess the signature fail to decode the transmitted message. The reputation of such a device degrades preventing them to participate in further communication. Therefore, the communication between the devices other than reputation management is also secured through an independent authentication method. The process of service authentication is illustrated in Fig. 4.

4. Performance evaluation

The performance of the proposed RRAC is evaluated using an IoT communication environment modeled in an opportunistic network simulator with 200 mobile wireless devices. The devices are interconnected with an open cloud for stored information (dataset and images) retrieval. Table 2 describes the detailed simulation setting for evaluation.

Table 2
Simulation setting.

Parameter	Configuration value
Wireless devices	200
Request size	64 bytes
Number of infrastructure units	18
Bandwidth	2 Mbps
Request expiry time	240 ms
Signature size	280 bits
Time slots	310

The consistency of the proposed RRAC is assessed using the metrics: detection time, service overhead, adversary detected, service loss ratio, false positive rate and misdetection ratio. The proposed method is compared with the existing sec Trust-RPL [27], MFTM [30] and FSDN-IDPS [20] for the above metrics.

4.1. Analysis of detection time

In Fig. 5, the comparative analysis of detection time between the existing methods and the proposed RRAC is illustrated. The process of detection is instantaneous for both communicating and idle devices in the proposed RRAC. The reputation of the device is decided using c_r and am that aids to select the appropriate resource or request processing neighbor. The SP updates the activity of the device in t_s to ensure the reliability of the communication. The reliability of the device is determined using R_i wherein a device failing to satisfy $\Delta\varphi_r < R_{th}$ or $\prod R_d > R_{th}$ is announced as malicious. This process of constraint verification is repeated within the time slot t_s for all d_r and d_{nr}. Therefore, the time taken for detecting a malicious device/user in the proposed RRAC is less. Besides, the detecting process is also extended for $(2TS - s_{cd})$ interval improving swift malicious detection.

4.2. Service overhead analysis

Fig. 6 portrays the service overhead comparison between the existing methods and proposed RRAC. The chances of services rediscovery and service disconnection are reduced in RRAC through internal and service security measures. If a device acts as a request handler then it must satisfy $\Delta\varphi_r < R_{th}$ and $\prod R_d > R_{th}$. Contrarily, if a device acts as a resource, then it must satisfy am (Eq. (19)). The above cases are required to retain the reliability of the devices during the activity verification time period. Therefore, the service discovering and responding routes are ensured to be secure with optimal neighbor/resource selection respectively. This ensures uninterrupted service discovery and responsive flows between the devices with an end-to-end security. Hence, the chances of service disconnection, malicious

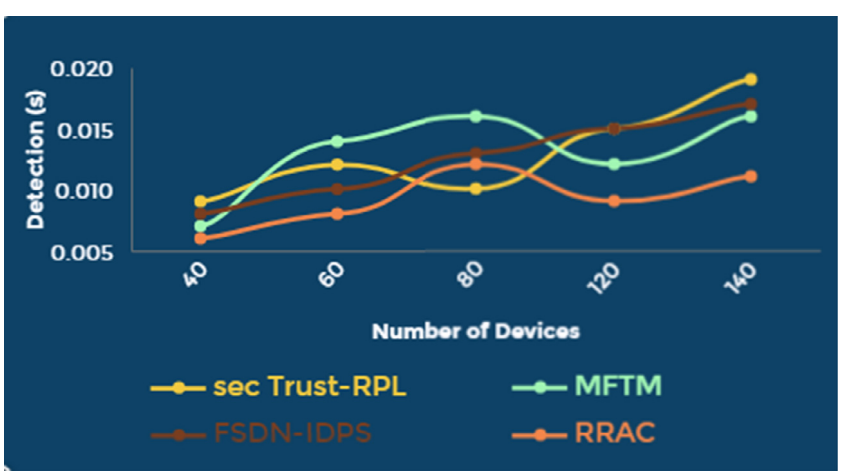

Fig. 5. Devices versus detection time.

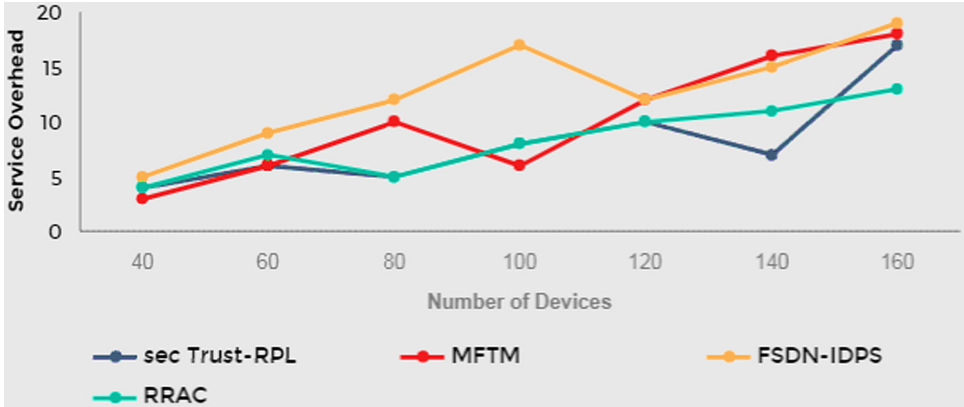

Fig. 6. Devices versus service overhead.

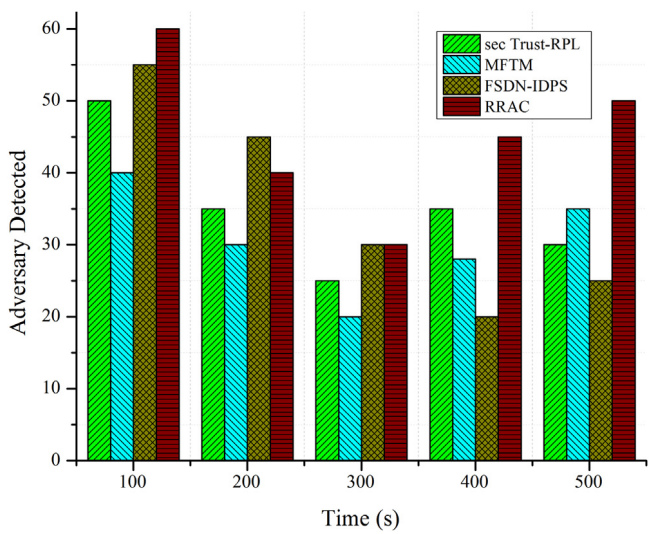

Fig. 7. Time versus adversary detected.

DoS are prohibited, requiring formal control requests. In the existing methods, the communication overhead is considered for rediscovery and disconnection of high routes.

4.3. Analysis of adversary detected

The prolonged operation time of the SP increases the rate of adversary detection. In the proposed RRAC, the operation time is split into time slots for updating the reputation of the devices. The number of verification instances within a specific t_s is dependent on d_r and d_{nr}; the number of updates for device reputation is high compared to the other methods. Therefore, the change in $\prod \mathrm{R_d}$ or $\Delta\varphi_\mathrm{r}$ of a device if found to unsatisfied the reliability constraints, then the reputation of the device at t is degraded and it is detected as malicious. The process of reputation management is augmented at the communicating time intervals t_p based on am handling. The diverse factors of reputation update increase with the operation time of the SP retaining the level of detection. The rate of detection varies with the increase in d_r or d_{nr}. The proposed RRAC eases reputation updates by estimating ΔE that précis's adversary detection (Fig. 7). On average, 62 adversaries are detected at different time interval.

4.4. Service loss analysis

The ratio of service loss for the detection adversary is compared between the existing and proposed methods in Fig. 8. The Constructive features of higher detecting, smaller detecting time and diverse reputation management improves the consistency of service communication. The consistency of the communication is achieved by selecting secure request handling neighbors and resource devices. These advantages help to improve seamless service availability and fewer failure rates. In the proposed method, both neighbor level and communication level security is ensured through reputation management and service authentication. Hence the changes for service falsification and DoS are detected at any instance of time t. The detection and reputation update of the devices are announced for the d_r periodically.

4.5. False positive rate analysis

Fig. 9 portrays the comparative analysis between the existing methods and the proposed RRAC for false-positive rate metric. In reputation management and update process, the observations and recommendations of d_{nr} and R_{nr} are considered. The rate of $\mathrm{R_{nr}}/(R_{nr}+R_r)$ from a single d or multiple devices increases the false positive rate observed in a particular time interval. The impact of d_{nr} and R_{nr} in the proposed RRAC is minimized by estimating Rth and $\Delta\varphi_\mathrm{r}$ based R_i. This helps to be minimized by the impact of R_{nr} from the same device and the number of non-reliable devices. The SP filters the reputation update of the unreliable devices at each t_s with the estimation of R_i and constraint verification. Though the un-reliable devices updated are not completely diminished, the proposed RRAC achieves a less false-positive rate compared to the existing methods.

4.6. Misdetection analysis

In Fig. 10 the misdetection ratio for the existing and proposed methods is analyzed. An increase in false alarm increases the misdetection rate. The reputation updates from d_{nr} increase the chances of misdetection due to the selfish behaviors of the devices. The proposed RRAC differentiates updates from d_{nr} and d_r by computing ΔE to improve the precision of reputation update. The neighbor's reputation is approximated with ΔE elimination at t_s. Besides the detection is extended in $(TS + s_{cd})$. Therefore the detection process re-estimates the reputation update irrespective of the t_s. For the extended certificate timestamp, the process is progressed with a new set of reputation values. However, at each t_s, the d_{nr} and R_{nr} are minimized to identify the precise reputation. This avoids the impacts of $R_{nr}/(R_{nr}+R_r)$ and $d_{nr}/(d_{nr}+d_r)$ minimizing the misdetection of the proposed RRAC.

5. Conclusion

In this manuscript role based reputation access control method is proposed for improving IoT communication reliability. Reliability is improved by providing communication and device-level security in this

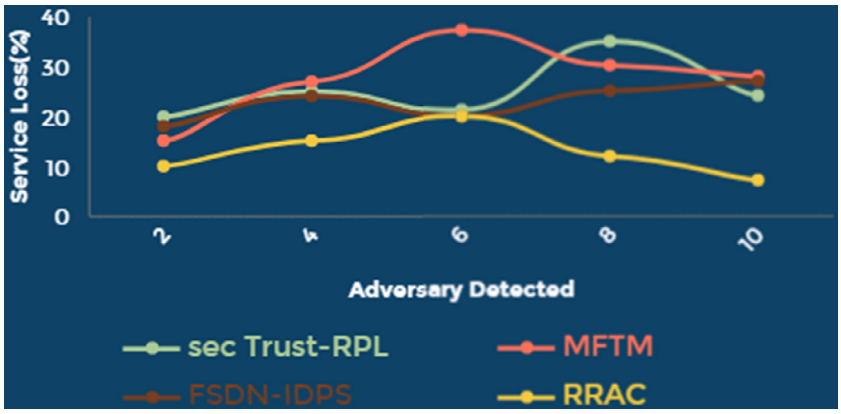

Fig. 8. Adversary detected versus service loss.

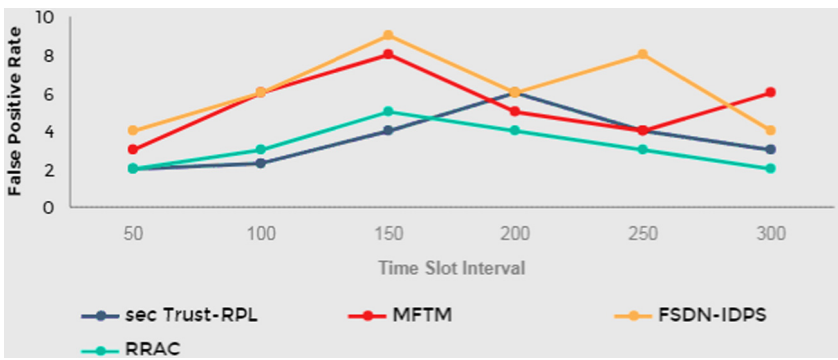

Fig. 9. Time slot intervals versus false positive rate.

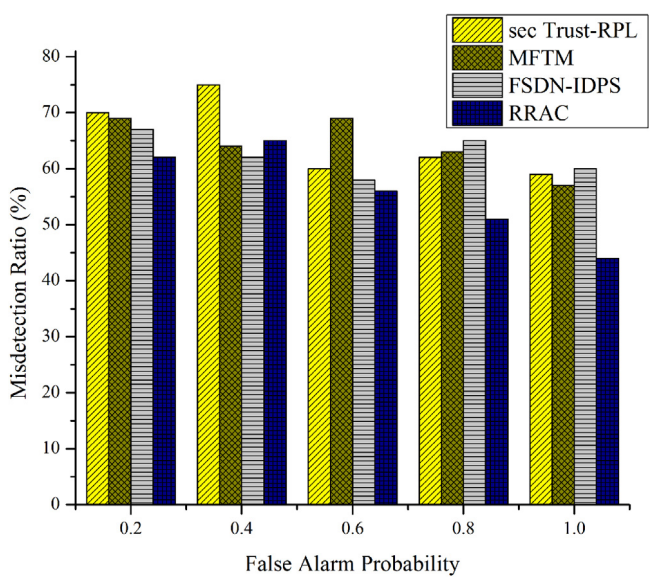

Fig. 10. False alarm probability versus misdetection ratio.

proposed method. The role of the IoT device is independently determined based on reputation estimated by the service provider (SP). The SP analyzes the communication characteristics of the device to estimate its reputation. The precision of reputation management is achieved by discarding unreliable devices based on false reputation information. Besides, communication security using linear hashing and hyperelliptic curve based digital signature ensures service authentication among the reputed devices. The SP updates the reputation of the device based on service authentication characteristics to improve the security level. Experimental results prove the consistency of the proposed method by improving adversary detection and minimizing detection time, service overhead and loss, false positive rate and misdetection ratio.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Mohammed Amoon: Conceptualization, Methodology, Writing - review & editing. **Torki Altameem:** Data curation, Formal analysis. **Ayman Altameem:** Validation, Visualization, Writing.

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